ANALIZA KODA ANDROID APLIKACIJA POMOĆU SPOTBUGS ALATA

- omogućeno je automatsko pokretanje SpotBugs alata na više projekata odjednom uz pomoć kreirane Java skripte “SpotBugs.java”

- skripta omogućava dodavanje određenog koda u *build.gradle* fajlove pomoću kojeg se kreira spotbugsMain task koji se izvrši pozivanjem komande *./gradlew spotbugsMain*. Pomoću te komande se kreiraju pojedinačni izvještaji za svaki projekat kojima se pristupa ili kroz foldere ili kroz finalni izvještaj (koji se također kreira) gdje pišu neke osnovne stvari o upozorenjima za taj pojedinačni projekat i koji se otvara odmah po završetku skripte
- skripta je puna komentara pa valjda neće biti problema u snalaženju

PREDUSLOVI:

- projekti moraju biti kreirani kao Android projekti i build-ani barem jednom
    - ovo je naznačeno zato što sam primijetila da su na GitHubu i Bitbucketu drugačije raspoređeni fajlovi nego kada se kreiraju kao Android projekti lokalno (ne znam zašto je tako)
    - uglavnom, ja sam testirala nad Android projektima koji su klonirani kroz okruženje i raspored je malo drugačiji tako da ova skripta za onaj raspored koji je na gitu vjerovatno neće raditi (pa sada sve zavisi kakav je raspored gdje, vidjećemo kada mi pošaljete projekte, sada sam testirala na nekim spiralama i vježbama od prošle godine)
    - pretpostavljam da će projekti biti kreirani kao Android projekti
- Andorid SDK mora biti promijenjen na Vaš SDK (ako već nije)
- Gradle verzija mora biti 5+, jer SpotBugs ne podržava niže

POKRETANJE:

- popuniti *paths.txt* sa putanjama na projekte koji se testiraju
    - putanje poredane jedna ispod druge
- pokrenuti u konzoli, iz direktorija gdje su skripta i *path.txt*:
    - javac SpotBugs.java
    - java SpotBugs
- omogućeno je pokretanje koliko god želimo puta, testira se da li je kod već dodan u *build.gradle*

SLUČAJEVI ŠTA JE SVE URAĐENO:

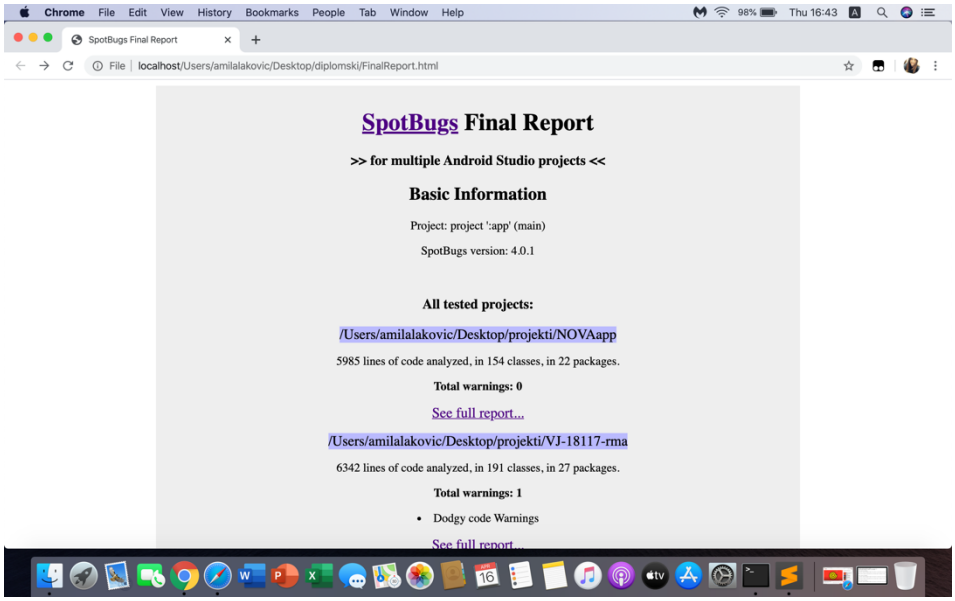
- **ako je sve od preduslova zadovoljeno trebalo bi ispravno raditi**
    - u konzoli se ispiše za svaki projekat da li je BUILD SUCCESSFUL

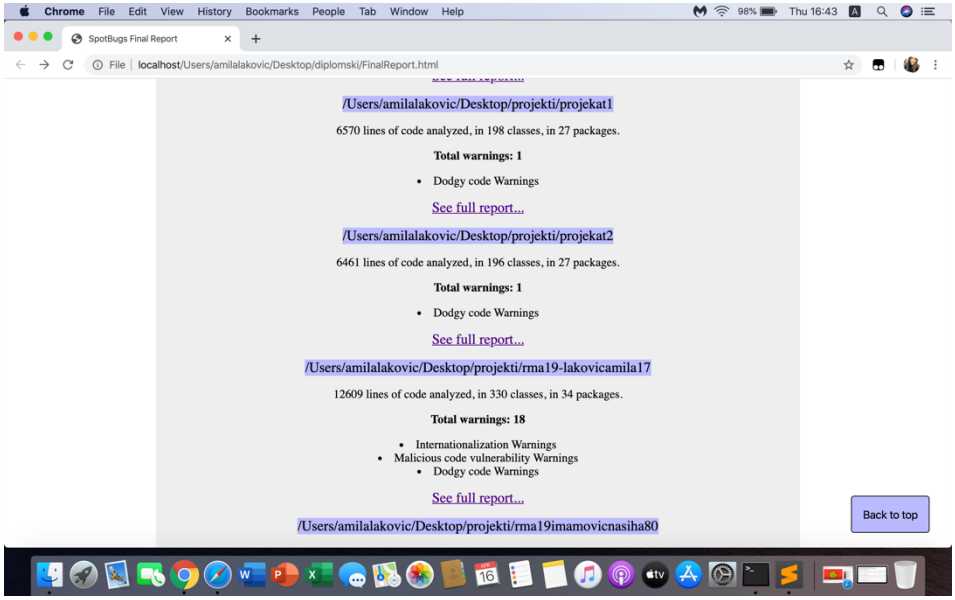
```
diplomski — -zsh — 97×49

Opening FinalReport.html...

amilalakovic@Amilas-MacBook-Pro diplomski % javac SpotBugs.java
amilalakovic@Amilas-MacBook-Pro diplomski % java SpotBugs
> Task :app:preBuild UP-TO-DATE
> Task :app:preDebugBuild UP-TO-DATE
> Task :app:checkDebugManifest UP-TO-DATE
> Task :app:generateDebugBuildConfig UP-TO-DATE
> Task :app:javaPreCompileDebug UP-TO-DATE
> Task :app:compileDebugAidl NO-SOURCE
> Task :app:compileDebugRenderscript NO-SOURCE
> Task :app:mainApkListPersistenceDebug UP-TO-DATE
> Task :app:generateDebugResValues UP-TO-DATE
> Task :app:generateDebugResources UP-TO-DATE
> Task :app:mergeDebugResources UP-TO-DATE
> Task :app:createDebugCompatibleScreenManifests UP-TO-DATE
> Task :app:processDebugManifest UP-TO-DATE
> Task :app:processDebugResources UP-TO-DATE
> Task :app:compileDebugJavaWithJavac UP-TO-DATE
> Task :app:compileDebugSources UP-TO-DATE
> Task :app:mergeDebugShaders UP-TO-DATE
> Task :app:compileDebugShaders UP-TO-DATE
> Task :app:generateDebugAssets UP-TO-DATE
> Task :app:mergeDebugAssets UP-TO-DATE
> Task :app:processDebugJavaRes NO-SOURCE
> Task :app:mergeDebugJavaResource UP-TO-DATE
> Task :app:checkDebugDuplicateClasses UP-TO-DATE
> Task :app:transformClassesWithDexBuilderForDebug UP-TO-DATE
> Task :app:mergeExtDexDebug UP-TO-DATE
> Task :app:mergeDexDebug UP-TO-DATE
> Task :app:validateSigningDebug UP-TO-DATE
> Task :app:signingConfigWriterDebug UP-TO-DATE
> Task :app:mergeDebugJniLibFolders UP-TO-DATE
> Task :app:mergeDebugNativeLibs UP-TO-DATE
> Task :app:stripDebugDebugSymbols UP-TO-DATE
> Task :app:packageDebug UP-TO-DATE
> Task :app:assembleDebug UP-TO-DATE
> Task :app:compileJava NO-SOURCE
> Task :app:processResources NO-SOURCE
> Task :app:classes UP-TO-DATE
> Task :app:spotbugsMain UP-TO-DATE

BUILD SUCCESSFUL in 2s
25 actionable tasks: 25 up-to-date
SUCCESSFULLY EXECUTED COMMAND, see if build is successful!!!

> Task :app:preBuild UP-TO-DATE
> Task :app:preDebugBuild UP-TO-DATE
```

    - također ima nekih ispisa na kraju, onako stavljeno…

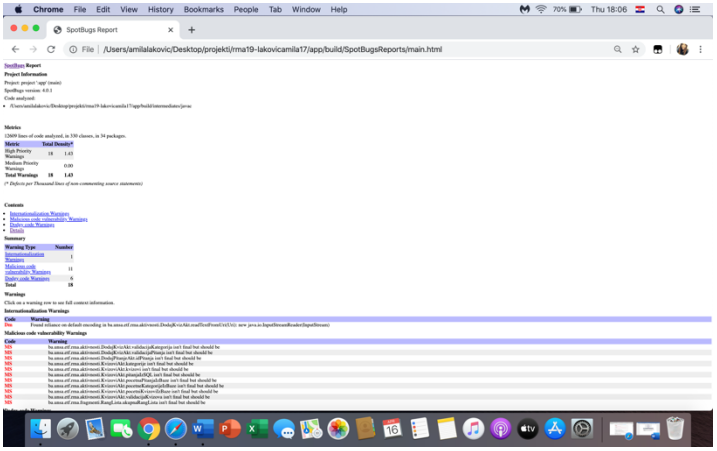
```
See https://docs.gradle.org/5.4.1/userguide/command_line_interface.html#sec:command_line_warnings

BUILD SUCCESSFUL in 0s
27 actionable tasks: 27 up-to-date
SUCCESSFULLY EXECUTED COMMAND, see if build is successful!!!

All individual project reports can be located in -> app -> build -> SpotBugsReports!!

File created: FinalReport.html

Opening FinalReport.html...

amilalakovic@Amilas-MacBook-Pro diplomski %
```

    - sada bi se trebao automatski kreirati (u tom direktoriju) i otvoriti finalni izvještaj za projekte koji su upisani u *paths.txt*. Taj izvještaj je izgleda ovako:



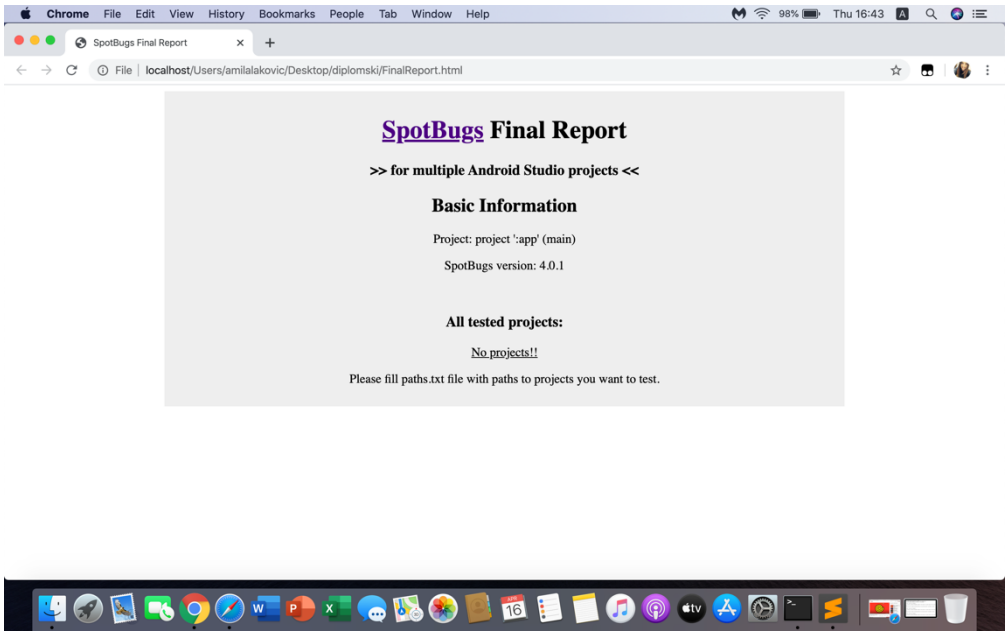


- pokušala sam napraviti da finalni izvještaj bude sličan originalnim izvještajima (zoom out-ovan je pa se ne vidi fino...)

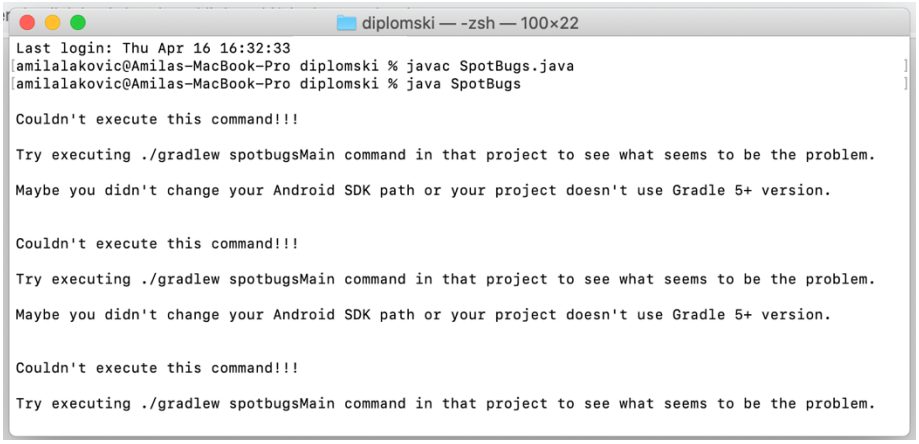


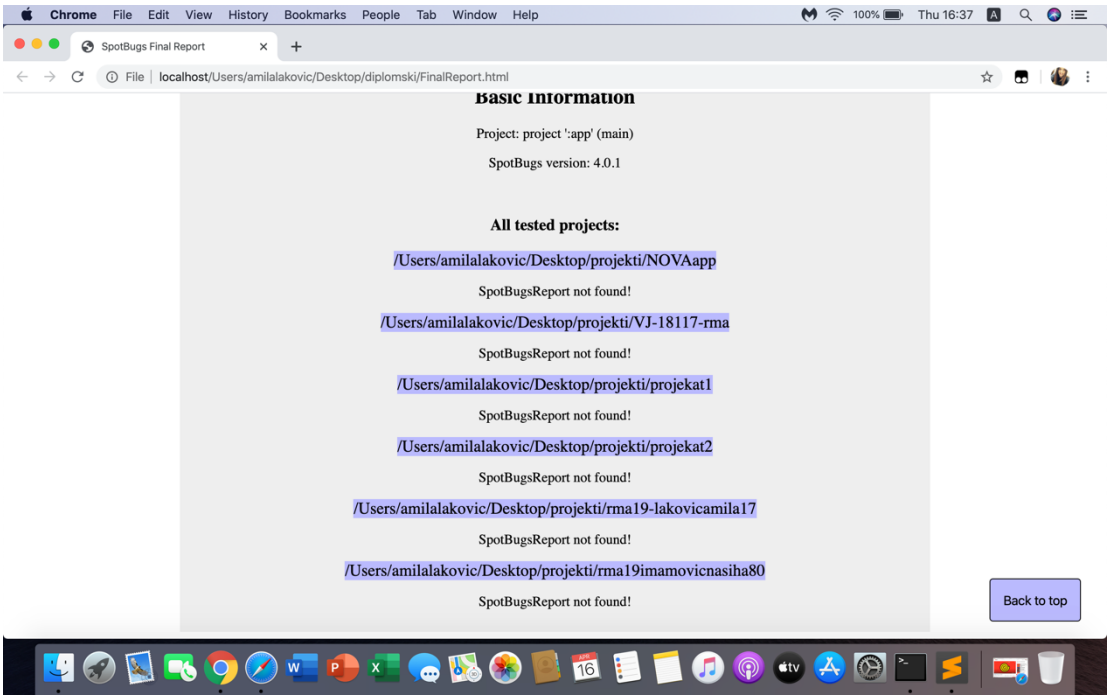
- u finalnom izvještaju sam zapisala samo neke bitne informacije iz pojedinačnih izvještaja, ali se kroz finalni može pristupiti čitavom tom izvještaju da se sve fino pregleda
- također sam dodala ovaj button “Back to top” koji se stvori kada se počne scrollati ekran
  - pošto pretpostavljam da će biti veliki broj projekata išla sam logikom da je lakše vratiti se preko buttona nego scrollati opet nazad

- **u slučaju da nešto od preduslova nije zadovoljeno, svi izuzeci se hvataju i izvještaj se kreira, samo malo drugačiji**
  - u slučaju da *paths.txt* ne postoji, ili nema ništa u tom fajlu izvještaj se kreira, samo malo drugačiji



  - slučaj kada nijedan od projekata ne zadovoljava preduslove
    - u tom slučaju nije moguće pokretanje komande za spotbugs i dolazi do greške, što bude ispisano u konzoli
    - ali izvještaj se i dalje kreira, sa malo drugačijim tekstom



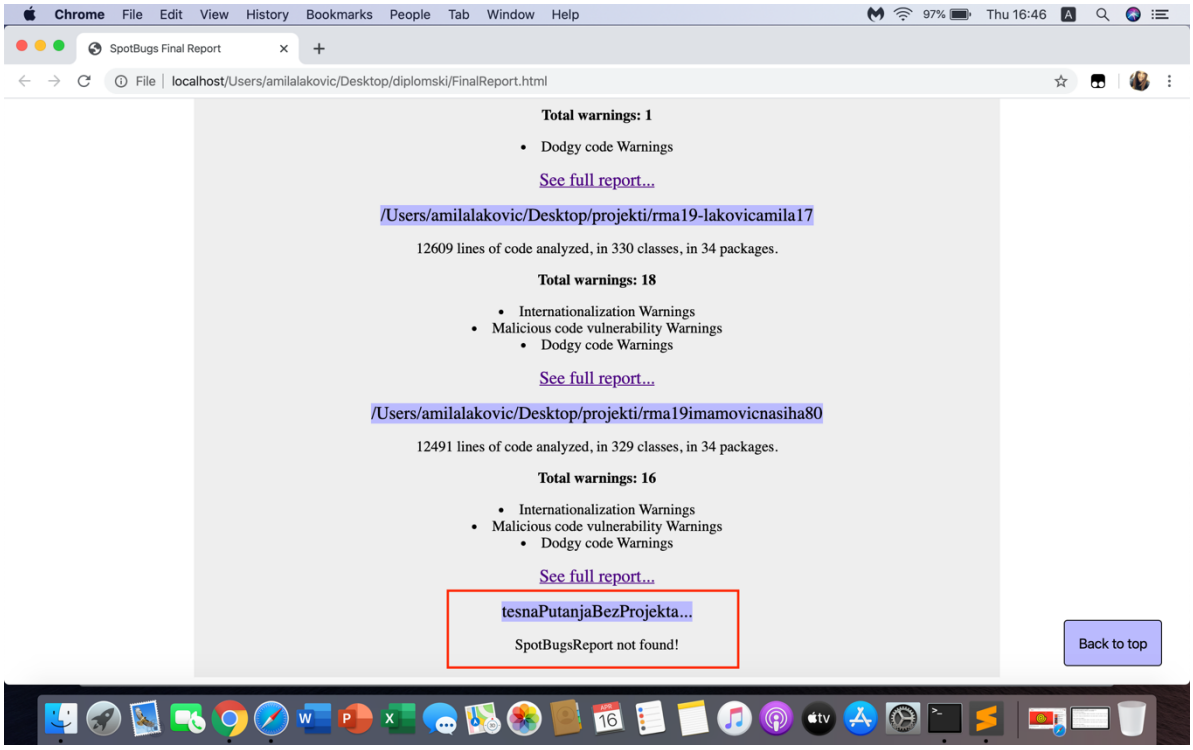


- prethodni slučaj, samo kada je samo jedan projekat neispravan
  - ovo je prvenstveno urađeno zbog ovakvih slučajeva, da se ne bi desilo da se prekine rad skripte ako samo jedan od projekata nije ispravan.. onda se može vidjeti u izvještaju koji je to projekat
  - na slici se vidi da je ispisana greška za prvi neispravni projekat, ali izvršavanje ostalih projekata se izvršava neometano
  - na izvještaju se samo ispiše da SpotBugsReport nije pronađen za taj projekat

```
diplomski — -zsh — 101×41
Last login: Thu Apr 16 16:36:45
amilalakovic@Amilas-MacBook-Pro diplomski % javac SpotBugs.java
amilalakovic@Amilas-MacBook-Pro diplomski % java SpotBugs

Couldn't execute this command!!!

Try executing ./gradlew spotbugsMain command in that project to see what seems to be the problem.

Maybe you didn't change your Android SDK path or your project doesn't use Gradle 5+ version.

> Task :app:preBuild UP-TO-DATE
> Task :app:preDebugBuild UP-TO-DATE
> Task :app:compileDebugAidl NO-SOURCE
> Task :app:compileDebugRenderscript NO-SOURCE
> Task :app:checkDebugManifest UP-TO-DATE
> Task :app:generateDebugBuildConfig UP-TO-DATE
> Task :app:javaPreCompileDebug UP-TO-DATE
> Task :app:mainApkListPersistenceDebug UP-TO-DATE
> Task :app:generateDebugResValues UP-TO-DATE
> Task :app:generateDebugResources UP-TO-DATE
> Task :app:mergeDebugResources UP-TO-DATE
> Task :app:createDebugCompatibleScreenManifests UP-TO-DATE
> Task :app:processDebugManifest UP-TO-DATE
> Task :app:processDebugResources UP-TO-DATE
> Task :app:compileDebugJavaWithJavac UP-TO-DATE
> Task :app:compileDebugSources UP-TO-DATE
> Task :app:mergeDebugShaders UP-TO-DATE
> Task :app:compileDebugShaders UP-TO-DATE
> Task :app:generateDebugAssets UP-TO-DATE
> Task :app:mergeDebugAssets UP-TO-DATE
> Task :app:processDebugJavaRes NO-SOURCE
> Task :app:mergeDebugJavaResource UP-TO-DATE
> Task :app:checkDebugDuplicateClasses UP-TO-DATE
> Task :app:mergeExtDexDebug UP-TO-DATE
> Task :app:transformClassesWithDexBuilderForDebug UP-TO-DATE
> Task :app:mergeDexDebug UP-TO-DATE
> Task :app:validateSigningDebug UP-TO-DATE
> Task :app:signingConfigWriterDebug UP-TO-DATE
> Task :app:mergeDebugJniLibFolders UP-TO-DATE
> Task :app:mergeDebugNativeLibs UP-TO-DATE
> Task :app:stripDebugDebugSymbols UP-TO-DATE
```

- također prethodni slučaj, jedan neispravan projekat
  - mislim da se to može najčešće desiti ako putanja do projekta nije ispravna, pa je ovdje to testirano na nekom stringu koji je dodan u *paths.txt* fajl

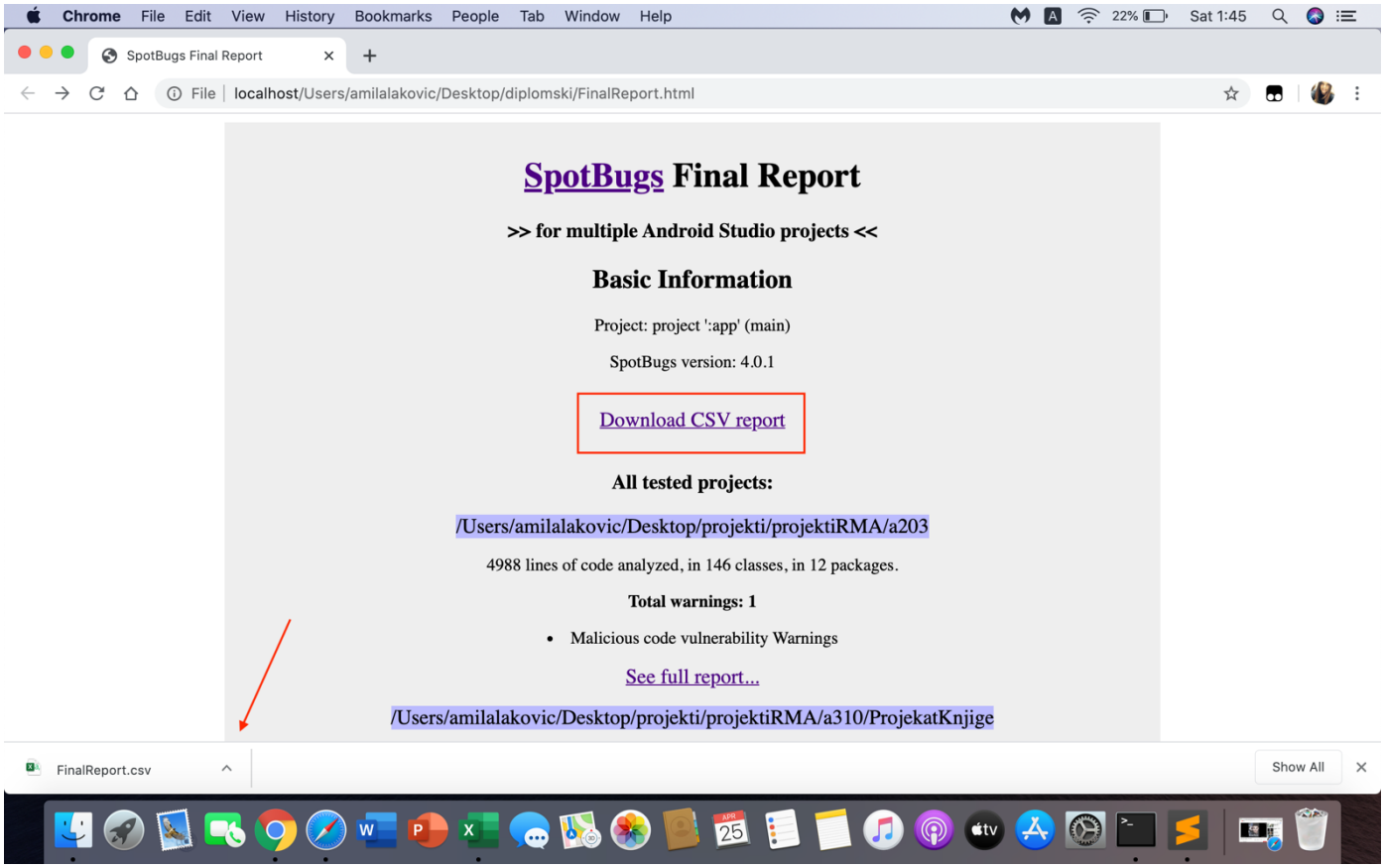
```
diplomski — -zsh — 80×24
BUILD SUCCESSFUL in 1s
27 actionable tasks: 27 up-to-date
SUCCESSFULLY EXECUTED COMMAND, see if build is successful!!!

java.io.FileNotFoundException: tesnaPutanjaBezProjekta.../build.gradle (No such
file or directory)
        at java.io.FileInputStream.open0(Native Method)
        at java.io.FileInputStream.open(FileInputStream.java:195)
        at java.io.FileInputStream.<init>(FileInputStream.java:138)
        at java.io.FileInputStream.<init>(FileInputStream.java:93)
        at java.io.FileReader.<init>(FileReader.java:58)
        at SpotBugs.SpotBugsInGradleExists(SpotBugs.java:102)
        at SpotBugs.insertTextInGradle(SpotBugs.java:79)
        at SpotBugs.main(SpotBugs.java:34)
java.io.FileNotFoundException: tesnaPutanjaBezProjekta.../build.gradle (No such
file or directory)
        at java.io.FileOutputStream.open0(Native Method)
        at java.io.FileOutputStream.open(FileOutputStream.java:270)
        at java.io.FileOutputStream.<init>(FileOutputStream.java:213)
        at java.io.FileWriter.<init>(FileWriter.java:107)
        at SpotBugs.addTextToFile(SpotBugs.java:125)
        at SpotBugs.insertTextInGradle(SpotBugs.java:80)
        at SpotBugs.main(SpotBugs.java:34)
java.io.FileNotFoundException: tesnaPutanjaBezProjekta.../app/build.gradle (No s
```



Nadam se da je ovo zadovoljavajuće.

UPDATE [CSV]

- dodana je opcija da se izvještaj kreira kao csv fajl
- urađeno je na način da sve što je do sada urađeno je ostalo isto, dakle i dalje se kreira html izvještaj i otvara pri završetku skripte
- promjena je što se sada kreira i csv fajl u istom direktoriju te mu se može pristupiti ručno ili kroz html izvještaj
- dodana je opcija na html izvještaju da se downloaduje csv fajl zbog lakšeg pristupa



- csv se otvara u Microsoft Excelu i ispisuje sve što ste tražili u zadnjem mailu
- obrađeni su isti izuzeci koji su već ranije navedeni

- kada ne dolazi do greške:

A	B	C	D	E	F
	Project	High Priority Warnings [HPW]	Medium Priority Warnings [MPW]	HPW/Average HPW	MPW/Average MPW
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a203	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a310/ProjekatKnjige	3	0	3.428571429	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a373	2	0	2.285714286	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a394	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a491	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a499	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a570/rmaspirala1	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a573/S1_17537	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a580	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a603/spiralasefer1	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a630/Spirala	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a667/Spirala1/Spirala12	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a703/Spirala1	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a720/Spirala17702NB	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a723	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a731	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a769	0	0	0	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a783	2	0	2.285714286	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a785/rma18corovic17758/Biblioteka	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a914/Biblioteka23	2	0	2.285714286	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a929/dinbostandzic17992	3	0	3.428571429	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a935/Spirala1	2	0	2.285714286	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a950	1	0	1.142857143	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a986/Knjige	0	0	0	CAN'T DIVIDE BY ZERO!!!
Total PW		21	0		
Average PW		0.875	0		

  - izabrala sam opciju da se u nazivu projekta ispisuje čitava putanja do projekta, kao što je i u htmlu
  - to je urađeno zato što sam vidjela da kada ste mi poslali projekte dosta njih ima dodatnih foldera unutar foldera itd.
  - *ako treba ovo da promijenim na samo nazive, recite… samo u tom slučaju da li da budu nazivi foldera koje ste mi poslali ili nazivi projekata?*

- dodala sam opciju da mi ispiše ako dođe do dijeljenja sa 0, da ne bi ispisivalo NaN
- također je dodan ispis zbira svih HPW i MPW na kraju svih projekata, te prosječan broj PW svih projekata kao zbir/broj projekata

- ovo je neki projekat bez warningsa testiran nekoliko puta i vidi se da se u obje kolone za PW ispisuje da se ne može dijeliti nulom

A	B	C	D	E	F
	Project	High Priority Warnings [HPW]	Medium Priority Warnings [MPW]	HPW/Average HPW	MPW/Average MPW
	/Users/amilalakovic/Desktop/projekti/NOVAapp	0	0	CAN'T DIVIDE BY ZERO!!!	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/NOVAapp	0	0	CAN'T DIVIDE BY ZERO!!!	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/NOVAapp	0	0	CAN'T DIVIDE BY ZERO!!!	CAN'T DIVIDE BY ZERO!!!
	/Users/amilalakovic/Desktop/projekti/NOVAapp	0	0	CAN'T DIVIDE BY ZERO!!!	CAN'T DIVIDE BY ZERO!!!
Total PW		0	0		
Average PW		0	0		

- dodan je i slučaj kada nema projekata u *paths.txt*, ispiše se isto kao i u html izvještaju

A	B	C	D	E	F	G
	Project	High Priority	Medium Pri	HPW/Averag	MPW/Average MPW	
	NO PROJECTS!!					
	Please fill paths.txt file with paths to projects you want to test.					

- također je i obrađen slučaj kada kod jednog projekta, ili više, dođe do neke greške pa se SpotBugs izvještaj ne kreira… ispiše se isto kao i u html izvještaju

A	B	C	D	E	F	G	H
	Project	High Priority	Medium Pri	HPW/Averag	MPW/Average MPW		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a203	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a310/ProjekatKnjige	3	0	3.78947368	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a373	/	/	/	/	SpotBugsReport not found!	
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a394	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a491	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a499	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a570/rmaspirala1	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a573/S1_17537	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a580	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a603/spiralasefer1	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a630/Spirala	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a667/Spirala1/Spirala12	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a703/Spirala1	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a720/Spirala17702NB	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a723	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a731	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a769	0	0	0	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a783	2	0	2.52631579	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a785/rma18corovic17758/Biblioteka	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a914/Biblioteka23	2	0	2.52631579	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a929/dinbostandzic17992	3	0	3.78947368	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a935/Spirala1	2	0	2.52631579	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a950	1	0	1.26315789	CAN'T DIVIDE BY ZERO!!!		
	/Users/amilalakovic/Desktop/projekti/projektiRMA/a986/Knjige	0	0	0	CAN'T DIVIDE BY ZERO!!!		
Total PW		19	0				
Average PW		0.79166667	0				